ChatGPT

Empirical Rule Calculation

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Source: https://chatgpt.com/c/698c1c29-6048-8327-bd91-c29989b90e98

If the mean life of the corona virus is 180 min and the standard deviation is 20 min. Then, 99.7% of all viruses could be active between _____ min and _____ min.

We can solve this using the **empirical rule (68-95-99.7 rule)** for a normal distribution.

The empirical rule says:

- About **68%** of data falls within ±1σ from the mean.
- About **95%** falls within ±2σ.
- About **99.7%** falls within ±3σ.

Here:

- Mean $\mu = 180$ min
- Standard deviation $\sigma = 20$ min
- We want the range for **99.7%**, which is ±3σ.

$$\text{Lower limit} = \mu - 3\sigma = 180 - 3(20) = 180 - 60 = 120$$

$$\text{Upper limit} = \mu + 3\sigma = 180 + 3(20) = 180 + 60 = 240$$

✅ So, **99.7% of all viruses could be active between 120 min and 240 min**.

If you want, I can also explain **why ±3σ covers almost all values** in simple words. Do you want me to?